BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: B.Tech. **SEMESTER : V**
BRANCH: AI **SESSION : MO/2025**

SUBJECT: AI363 INTRODUCTION TO MACHINE LEARNING

TIME: 02 Hours **FULL MARKS: 25**

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
***4*. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates**

			CO	BL
Q.1(a)	Define Machine Learning. Explain its significance with real-life examples.	[2]	1	1,2
Q.1(b)	Differentiate between **Supervised**, **Semi-supervised**, and **Unsupervised** learning with suitable examples.	[3]	1	2,4
Q.2(a)	Explain the concept of **Eigenvalues and Eigenvectors**. Show step-by-step calculation of eigenvalues for the matrix $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$	[2]	2	2,3
Q.2(b)	Describe the working of **Principal Component Analysis (PCA)**. How does PCA help in dimensionality reduction?	[3]	2	2,3
Q.3(a)	In linear regression, how is prediction accuracy measured ?	[2]	1	
Q.3(b)	Apply the technique of linear regression (closed form solution) on the following data:	[3]	3	

X	1	2	34	4	5
y	12	18	22	28	35

Compute the value of y for x = 8.

Q.4(a)	Discuss the **Bias-Variance Trade-off** in supervised learning. Illustrate with a suitable diagram.	[2]	2	2,4
Q.4(b)	Explain **Regularization** in linear regression. How does it help to reduce overfitting?	[3]	2	2,5
Q.5(a)	Explain **Logistic Regression** as a classification algorithm. Derive its cost function.	[2]	3	2,4
Q.5(b)	State important issues in classification problems. Suggest possible solutions.	[3]	3	4

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